COMPARISON: ALIGNING CAPACITOR SELECTION WITH THE LATEST AUTOMOTIVE DEMANDS

Name of Company	Name of Series	Type	Capacitance Range	Operational Temperature	ESR Value (mΩ)	Notable Features	Key Application	Compliance and Qualification
KEMET	A768	Solid Polymer Aluminum Capacitor	18µF up to 1000µF	−55°C to +125°C	15 to 54 (@ 100kHz, 20°C)	Taller base and extra terminals for physical support	Vehicular electrical subsystems, engine control units, and automotive DC/DC converters	RoHS, REACH, AEC-Q200
KEMET	FT-CAP	MLCC	180pF to 22µF	−55°C to +125°C	N/A	Conductive silver epoxy layer situated between the base metal and nickel barrier layers	Critical timing, tuning, circuits requiring low loss, circuits with pulse, circuits subject to high levels of board flexure or temperature cycling	RoHS, REACH, AEC-Q200
KYOCERA AVX	TCQ	Polymer	2.2µF to 470µF	-55 to +125°C	25 to 150 (@ 100kHz)	Conductive Polymer Electrode and a Benign Failure Mode Under Recommended Use Conditions	DC/DC converters, Telecommunication (coupling/decoupling), Automotive (body electronics, cabin controls, infotainment, comfort, after market etc)	RoHS, AEC-Q200
KYOCERA AVX	TCO	Polymer	10µF to 33µF	-55 to +150°C	100 to 150 (@ 100kHz)	Conductive Polymer Electrode and a Benign Failure Mode Under Recommended Use Conditions. Suited for higher temperature applications	DC/DC converters, Telecommunication (coupling/decoupling), Automotive (body electronics, cabin controls, infotainment, comfort, after market etc)	RoHS, AEC-Q200
KYOCERA AVX	Flexiterm Products	MLCC	220pF to 100nF	-55°C to +125°C	N/A	The nickel-plated copper terminations are supplemented by a proprietary layer that absorbs incident mechanical stresses	High Flexure Stress Circuit Boards, All kinds of engine sensors, Variable Temperature Applications	RoHS, AEC-Q200
Panasonic	EEH-ZS	Hybrid	47µF to 470µF	-55°C to +125°C	20mΩ to 100mΩ (@ 100kHz)	Auxiliary side terminals. Vibration Variants Can Withstand Shocks Of As Much As 30G	DC/DC Converters, AC/DC Converter, Under-The-Hood Applications (125°C)	RoHS, REACH, AEC-Q200

capacitors whose electrolyte is actually a liquid—a conductive solution—polymer capacitors have a solid electrolyte that's made up of conductive polymers. They are therefore more compact and there is no chance of leakage or the capacitor drying out. These components will have lower ESR values than MLCCs, with better performance across a wide range of operating conditions. However, there are increased costs involved.

A great example for this category would be Kemet's solid polymer aluminium capacitor. It is the solid polymer that makes these capacitors vibration-resistant. In their A768 series of solid polymer capacitors, a few design changes were made to increase their vibration tolerance even further. They were provided with a taller base

Fig. 4: TCQ automotive conductive polymer chip capacitor (Credit: AVX)

for better physical support. What's interesting about these capacitors is that they have extra terminals that do not provide electrical conductivity but are actually used to give the capacitor a stable connection to the PCB.

Kyocera AVX has two special series of automotive polymer capacitors: the TCQ series and the TCO series. TCQ is used for standard

Fig. 5: Panasonic EEH-ZK hybrid capacitor (Credit: Panasonic)

temperature conditions while TCO is used for applications that demand higher temperatures. Both of these meet all the 'AEC-Q200 Stress Test Qualification for Passive Components' requirements and are reliable for a wide range of temperatures. They are robust and provide stable